

ActInf Livestream #001.1 ~ "Narrative as active inference"

Livestream | 1:28:16

hello everyone

welcome to team com podcast one

tentatively titled really looking

forward

to this discussion i'm daniel and

along with me are yvonne alex do you

want to introduce yourself briefly

yeah my name is alex i just can't say

what

i'm a researcher in system management

school in moscow

by now just for protocol

hello my name is ivan i'm just

interested in internet active inference

stuff great well

to be interested in active inference is

all that's needed

for this paper and this discussion so

let's

get right into it we're gonna

structure the discussion like this first

we'll start with some warm-up

questions then we'll get to the paper

itself

and this isn't the end-all be-all

format for paper discussions but a few

big areas that we'll hit upon

we'll ask what are the big questions are

this paper what do the authors want to

address

then we'll go through the abstract line

by line because it's really important

to understand how the authors summarize

their own work

we'll look at the road map of the paper

at which sections come in which order
so that we can have a big bird's eye
perspective
on where the paper is going then
we'll go down into some definitions and
claims that the authors make
we can ask whether we agree with their
definitions or whether their claims
resonate with how we think about the
world then we'll have time
to have any questions that we want to
address
that are about the paper and then also
there's a few slides at the end where we
prepared some visualizations
about markov blankets and about active
inference itself
so let's get to the warm-up questions
and so
either of you can maybe give a first
thought here what what is a narrative
and how is a narrative different than a
story
and this can come from the free energy
active inference perspective or not
but one of you two guys go ahead
uh at the moment uh
i believe what first we what we need to
do what we need to define with
difference because uh if we have
two words uh it would mean something
different
but as in any case it's
depends on perspective and interest
from how we see it and um
[Music]
maybe his uh narrative and
ontology is also some different and

ontology is
different from the story as well so
some kind of more formal
definition for it we have a
great review
on different papers presented
in the text and so it
looks like as state of the art
in this field from scientific point of
view
but we if we go more engineering way and
trying to find
a way to and approaches to use
functions of his narrative
scientologists
and lately
stories very nice
and we'll get to see exactly their
definition
of narrative it's actually the first
sentences of the paper
uh yvonne anything there no
cool another question that is
sort of a fundamental in communication
but also
very important applied question is how
individual and collective narratives
interact
so do groups genuinely have narratives
or do group narratives just consist of
individual narratives
we can i mean we can move past these
these are really
just for warming up and thinking about
what the paper is arising out of and so
just the last
warm-up question that anyone could like
a stretch

before hitting the paper before you run
the paper you might want to stretch with
these questions
and just to really get us thinking
big about
what this paper could be about and what
science can tell us about narratives
this question is what does a functional
evolutionary or statistical perspective
on narratives
look like so instead of just approaching
it from a psychological
or from an anthropological perspective
how can we
take it deeper and connect it to other
areas
any last warm-up thoughts or we can just
go right into the paper
uh maybe
yes and it's about introduction
and first sentence in introduction um
about narratives a report of real or
imagined events and maybe i was not so
close inside but
maybe we need to define some
definition and perspective about events
what is it events for us at
this moment and in
in systems uh and events in uh
active inference what could be
here like a fundamental for us
good question and we're going to return
to their exact
definition of narrative we'll have it up
on a later slide
so keep that thought around and let's
just go into the paper
so the paper we're going to be

discussing

is called narrative as active inference
an integrative account of the functions
of narratives

and it was posted on psi archive which
is a pre-print server

the version we're looking at is from
june 18 2020

and the first author's name is nabil
buzagarin apologies for pronunciation
and uh there are several other authors
as well

the paper really sets out to address a
few

big questions the first question

uh is what does the concept of active
inference offer us

in the context of narrative or when
discussing the concept of narrative
that's the big question and then to take
it down

one more level they're going to
specifically ask how does this active
inference perspective

on narrative address questions related
to identity

episodic memory future projection event
segmentation

which is actually i trimmed it but it's
related to what you're discussing
storytelling enculturation and more

so let's get to the abstract um it's
kind of simulating

reading a paper you know i start from a
title then we'll go through the abstract
and then we'll go into the definitions
next but let's just go

bit by bit through the abstract so we

can really see what the authors think
they're doing in this paper
first sentence is basically saying that
narratives are important
and other people have recognized it
the second sentence says that research
has identified
several functions of narratives that are
important
uh if anyone has any thoughts that
they
want to add on to one of these sentences
or disagreement
they can uh also use the raise hand
feature in jitsi
so on the bottom left the hand if you
want to uh
say something um the abstract is not
too contentious in this paper
this is the key signal phrase that's
marking what they're actually doing in
the paper they're going to characterize
social and cognitive functions those are
the plurality of functions
of narratives in terms of the framework
of active inference
which we're going to talk a lot more
about in detail what is active inference
and then they say that active inference
is
depicting the fundamental tendency of
organisms to adapt
update and maintain inference about
their environment that's sort of the
beginnings of a definition of active
inference which we're going to come back
to
they then review literature related to

the function of narratives in all of those areas that we discussed identity event segmentation episodic memory etc and then the last sentence of the short abstract is we recast these functions of narratives in terms of active inference outlining a parsimonious model that can guide future developments in narrative theory research and clinical applications and so this is kind of a common theme that we see in active inference and free energy principle related research at this current moment in time the second half of the 2010s and the 2020 period which is um the authors usually consisting of a team with some disciplinary experts as well as some more free energy or active inference experts will approach an area like narratives or depression or evolution and they'll seek to integrate multiple phenomena under a common model so instead of saying well you got the short-term memory system and the long-term memory and the declarative memory and this you know this type of memory they'll sort of recast a lot of previous models into one common framework so that's what active inference can do

and the idea would be that by having an integrated framework we're going to get multiple benefits the first is that we have an elegant and a parsimonious way for addressing the otherwise unrelated phenomena arising from the same physical system so we shouldn't have 10 models for describing 10 different kinds of memory in the brain we should have one model for memory in the brain and then different aspects of it could relate to these phenomena that we want to actually explain and the second thing so it gives internal coherence to the study of systems to use an integrated framework but also it provides external coherence because now the information or the thermodynamic components of the brain can be considered alongside something like a computer so that's the abstract here's the road map of the paper and i just copied out the section titles and this is just so that even if some of these section titles sound a little bit like they're not fully specified or anything this is just so we can first sketch out the whole arc of the paper the first section is called the function of narrative and you can see they're going to be talking about narrative from a non-free energy

non-active inference perspective
they're going to be drawing on other
research for example
relating narrative to identity meaning
making which is also sometimes called
sense making
and coherence as well as storytelling
and they'll talk about the narrative
practice hypothesis which
is from social psychology the second
section of the paper
is explicitly about active inference
where they define
active inference and then they
specifically relate how active inference
can be thought of when looking at human
social and cultural life
and then the third section of the paper
no surprise
is narratives as active inference and so
here is where they're going to combine
section
one and two into discussing
how framing narrative as active
inference
gives us these features of a model that
we'd like to see
like it's elegant it's parsimonious it
has high explanatory power
it might lead to unique predictions
unique explanations
uh unique experiments that we could test
these are all the things that we care
about from a scientific
and a meta scientific perspective so
that's the big road map of the paper and
now we can
finally get into the first two sentences

of the paper
which are right here and alex this is
where you were talking about the
definition
of narrative so it's great that they
begin the paper
just with the definition of narrative
and i'll just read it out and then
either of you or both of you i'd love to
hear your thoughts where does this
resonate with you
where does it not resonate they write
narratives are reports of real or
imagined events
which can be presented in language
either verbally or textually
or through sequences of images or other
symbols
that's their first short definition and
the second sentence is
the narrativity of these reports lies
not just in their content
which may point directly to specific
events but also in their structure
that is the ways in which the linguistic
or other symbols convey a set of
relationships among events
especially temporal relationships so
what do you guys think
here for me what is important here that
uh its present idea
about uh
ability uh in a language to
present
to represent maybe to a person firstly
and he can share its uh in texts and
describe himself and maybe try to
understand

somebody using these symbols
yep yep yvonne anything
no not right now sure so i think uh
there's a few other words that we might
want to
also define and this is the nature
of language right we could always go and
ask well what is a report
is the narrative only real if it's
reported or what's the difference
between real and imagined
or what do you mean symbols so there's a
lot of
play in these definitions still but the
features that i
found interesting the first part is
that narratives are specific reports of
real or imagined events
so it's kind of like if a tree falls in
a forest and
no one hears it did it happen uh if no
one reports the narrative
there is no narrative okay if nobody's
speaking there's no narrator
so a narrative is a declarative event
in that telling the narrative or
experiencing it
is what makes it real and then also
this idea that a narrative is a sequence
of symbols
whether they're verbal symbols or text
symbols these symbols are representative
as symbols are and to bring it to the
second
section the string of symbols
it's not just that there's a shannon
entropy of the string or there's some
surprise or there's some sequence

it's actually that the representative nature of the symbols is inferring something about specific events again either real or imagined also how the events are linked and this is especially important in the context of things happening through time so that's their definition of narrative and

pretty shortly after that we get to one of their fundamental claims and there's going to be

a key sentence here that is actually meta scientific in the sense this is their evaluation of the research

literature they say that narrative has socio-interactional and cognitive representational functions however the extant literature tends to focus on only one of these two aspects

so in other words they're saying that the narrative has socio-interactional roles

for example the narrative of a country is often manifested through politics or through politics

uh or through policy and so there are social implications for the interactions between different narratives

and then that sort of bodies interacting with each other

but then inside of the head there's also this cognitive representational component of narrative

and that has to do with how different people experience narratives or how

different altered states of
consciousness
influence the ability to generate
narrative and then they write
in this paper we propose a unified model
that accounts for both of the functions
of narrative
so this is where they're taking stock
of the previous state of the literature
they're saying look
narrative literature falls into
basically either one of two categories
either it's about treating the person as
a black box
and then looking at how narratives occur
in the social context
or it goes the other approach which is
it zooms in on the brain or on the mind
from a neuro or from a psychological
perspective
and that ends up giving
not enough emphasis to the social
components of narrative
so my questions for you two are what are
the implications
of this unresolved dichotomy and what
might an answer to this problem look
like in other words what would it look
like to have a satisfying
science of narratives and what are the
implications
of us not having that satisfying science
of narratives
uh if there will be
an implication for one common model
and it will be possible to
predict outcomes
for testing this model in different

applications this could be more
scientifically
or at least engineering
yep and another engineering perspective
on this would be
false positive and false negative so
let's just say that we have
two different methods for dealing with
narrative one of them
doesn't give enough emphasis to what's
happening inside of the head
that's the socio-interactional one and
then the other
perspective on narrative doesn't give
enough
emphasis on what's happening outside of
the head and so we're going to end up
with false positives and false negatives
we're going to end up with lost in
translation we're going to end up with
situations
where one approach leads to one
prediction but then a different approach
leads to a different prediction
and without some sort of higher level
theory to help us resolve that
there's just not going to be clarity so
we can
see that having this unresolved
dichotomy not just is it theoretically
unsatisfying
uh it'd be like if you had one model for
physiology that happened you know just
in the liver
and one and outside of the liver but uh
that would be theoretically unsatisfying
but also it might lead to
some very real consequences when models

don't agree

and then people end up acting off of the
bad information

so no surprise here but they're going to
resolve this dichotomy

with active inference so here's how they
define

active inference and then we're going to
spend a lot of time

talking about active inference and
adding in some more figures because
there aren't any figures in this paper
which is totally fine

um but the crux of the issue
is about applying active inference to
the idea of narrative so we really want
to understand what active inference is
and then we'll use the active inference
framework to potentially reduce
our uncertainty about what narratives
are so they write

active inference is a new theory that
provides an account of the nature
and mechanisms of knowledge driven
context appropriate action
originally proposed as a theory of the
structure function and dynamics of the
brain

in the 2010 first in paper active
inference provides

a broad framework that has been extended
to explain living organisms and
cognitive systems across multiple scales
so that's the definition of active
inference and i i just tried a little
first

stab there's two big key
terms and they come up again and again

in this discussion and in others and
that's active inference
which is i put it in idea cloud because
this is the big idea
that we're talking about and on the
bottom right there is the free energy
principle
and the free energy principle is the
background
it's the uh it's the model that we're
working
under that we get active inference from
and in the 2018 interview
where fristen says that the free energy
principle is beyond
falsification it's at the axiomatic
level
so that doesn't mean it's indubitably
correct in fact axioms exist
beyond the space of correct versus
incorrect because correct and incorrect
only exist
within the context of an axiomatic set
and first
compares free energy principle to
something more like the principle of
least action
from physics in the sense that other
patterns
are fit within that so it's kind of like
the it's the canvas that we're going to
be drawing on
and so here are a few of the key things
that combine
to give us active inference one of them
is
informational thermodynamics another
important

topic is informational foraging and the
that's the active part is the actual
foraging for information
then another term that comes into play
is deep models so temporally deep models
with a
deep past and a deep future predictive
component
as well as you could almost call them
thick models so models that include
multiple modalities or multiple
perspectives
there's an element of the integration of
ecology evolution and development
so we're not looking to make a science
of narratives that only applies to
tweets or only applies to novels we want
something that's
fully grounded in ecology evolution and
development
and then also there's this key term the
markov blanket which we're going to go
into a little bit
more detail in the coming slides but
first off
what do you think about that what is
that uh
what does that make you guys think about
great overview about domains
and basic stones
for his ideas
so really as free energy principle the
principle
gave us opportunity
freedom to try to implement it to
explain different uh
systems and organisms so and
active inference as a more focused

on his domain around
the brain about how its work on
cognitive and metacognitive levels
and so on yep
and that's the key difference between
the
axiomatic principles which cannot be
falsified
in other words there's no observation
that would falsify
the free energy principle whereas
evolution by natural selection
just to give an example uh it could be
falsified
if we found some sort of uh anomalous
explanation or an anomalous observation
then
it could reduce the likelihood that a
certain model
is true in other words specific
hypotheses arise
from these other fields that i have
written in the black or the dashed line
like if you find that the organism is
only looking
one second into the future it's not a
deep model
so you can test whether it's a deep
model or not you can test the role of
ecology evolution development etc
you could look at whether the actions of
the organism are consistent with
thermodynamics you know if more energy
is coming out of the system than coming
in
then it violates thermodynamics so
that's something specific
and on the other hand free energy

principle is a little bit different
and uh there's so much to this and you
know every
every podcast we do every discussion we
have
we could spend so long discussing the
fundamentals
and so i feel like it's a good approach
each time it's like a coat of paint
we're just to keep on coming over the
same topics again and again
and going into them a little bit more
deeply each time
so let's look a little bit at the markov
blanket and here
i'm drawing from a paper of gallagher
and allen 2016 which is in the synthesis
the journal
and this paper is pretty nice because
it's not by fristin
uh it's a rarity you know but it's not
by fristen
but it contextualizes the work of
fristin and collaborators
from a philosophical perspective so here
are philosophers trying to talk about
markhaw blankets
they're focusing on this center node a
and they write the circle that's shaded
in gray the big one
represents the markov blanket of node a
this consists of a
its children parents and parents of
children
so here is how i'm thinking about this
pedigree
like a family tree is a metaphor for the
generative process that leads to

people or leads to a person you know
that any person
has to have two parents and if any
children there has to be a partner so
that's like s
in this the parent of child so
now what what's interesting to note is
in a world where the rules are different
for example
if three people were required to create
a child or we were able to create
a cloning technology where only one
person was required to create a
another human then the rules of this
system would be different and so the
mark called blanket would be drawn
differently
so generalize this idea
of a structural form or a scheme a
schema
for causal forces so it's like the two
parents cause
a now this is not saying oh those are
the only two things that cause a person
to exist
of course a civilization is required and
all these other things
but just within the context of a
pedigree
the all the only information you need to
know is the parents
and then there's times where
these connections that are being
extracted are statistical and there's
times where they're causal and this is
sort of um an interesting point
because it really gets at the heart of
what do these lines represent

and so there's times where the real world system we want we don't know about where the edges are and we want to use observed data to get the edges so we have a bunch of stock prices and we want to run a model that says okay when gold goes up then we want an arrow from the gold node to some other node for when the price responds in relationship to gold so that's like a statistical or numerical extraction of these edges and that is uh related to approaches like structural equation modeling sem modeling or causal entropic uh type of causal entropic forces and then there's other times like in the case of this graph where the relationships are genuinely causal so that's where the edge is actually representing a cause effect relationship of the states of the world so i'm going to nuance this marcotte blanket drawing a little bit so here it says in small script the subpartition of internal and external states is going to occur according to the free energy principle this is the caption of their figure and then i just added extra colors instead of just having the subscript so i blacked out the hidden external states

to really emphasize that the things that
are outside of the markov blankets
are truly speculative causes
so in this case it's like we don't know
who the grandparents of
a are we know that something led to the
parents
but we truly don't know what they are
and then we know that something has to
lead to the parents
of the child but again we don't know
where they are and then there could be
downstream effects of the child but we
don't know what they are
and so a like the focal individual
what's happening inside of that green
node is uh the internal states of a
the system and focus the system of
interest
and then you can think of the things
that a
causes whether by itself or with a
collaborator
are like the active the active states
those are the emitted actions of a
and then everything that is coming into
the markov blanket
but isn't an internal state and isn't an
active state
is a sensory state and it's kind of
weird to think about a pedigree as
having
sensory states but we'll see a few other
versions of markov blankets where
the term sense is going to be more
appropriate but the point is whether you
call it a sensory state or
just a markov blanket input it's the

same structure

any questions on that or there's some more visualizations of markov blankets yeah maybe let's um see some more cool cool so another key point about markov blankets is that they can be multi-scale

and so here i've borrowed a figure from ramstead at all 2018

and this is the answering schrodinger's question paper

which is a extremely powerful multi-level phrasing

of the free energy principle and so i've carried over those same hidden internal active and sensory states

idea and then added this uh pink or magenta color to the outside

and that's the markov blanket so we can see the hidden states are outside of the blanket

and then the active states the blue circles

are what penetrate outward those are actions that leave the markov blanket and then the red states are in incoming sensory data now the reason

we can talk about multi-scale markov blankets

is reflected here so this whole thing let's just say this is like a cell you know inside the cell you have some stuff going on and then you have actions coming out and you have sense coming in but that cell is actually also in the context of a larger

uh unit let's say a tissue and in this case
a bunch of cells are next to each other
and so each cell is doing
this at the cellular receptor level but
then
the organ can also be thought of as
having
internal external active and sensory
states
and what the mathematical claim
is is that there's a continuity
between these two different levels of
analysis
and so there's a few ways you can think
about this continuity
one is just the compositionality
argument
which is like okay if tissues are
composed of cells and cells are composed
of organelles
it's sort of a markov blankets all the
way down
approach but there's also some formal
continuities
as well as some possible ways which we
can
make a model of this level which is a
mixture of martial blanket states
and then here at the uh more granular
level we can actually
parameterize those relationships
and another way that we can look at the
mark call blanket again i'm
invoking the um previous magenta
uh uh pedigree version so here's the
pedigree version of the markov blanket
around

a the focal individual and then here
i've used the same
colors on a neural uh focused markov
blanket
so here the markov blanket is around the
brain or the brain of the body
and the internal states are what are
happening inside the brain
action states leave the brain and
sensations enter the brain
and then the hidden states are what are
happening outside of the brain
and then i put this kind of arrow in
quotation marks and this is really where
we get to the idea of a causal
generative model of the world and that's
when the brain
speculates about how causes in the world
might actually be related and it turns
out that
having a causal model of the world is
one of the most effective ways to reduce
your uncertainty
because if you're always just waiting
for sensations to show up
and then trying to deal with them by
changing your model or by doing actions
it's going to be a very confusing world
but if you can have a causal model of
the world that says for example
oh a car is something that emits a
certain light
and sound and heat and so none of those
different stimuli are surprising
because i have a generative model of the
world where a car is driving in the
street in front of me
and so that's what explains the

sensations you don't need to explain the
sensations of the sound
by changing your role model uh or by
doing action per se just making this uh
causal link in the world can reduce your
uncertainty a lot
but that's something that whether or not
it's in the world it's actually being
inferred by the brain
and then just to unpack the math
one more level and also connect the
brain out to
other systems here's a very similar
drawing
and this is just showing how um again
the internal states of the brain
influence the active states
which influence the world plus the world
has its own dynamics
those relate to sensory breaths which
influence the brain's internal model
and again that plays back into action
and similarly in the context
of a non-brain cognitive entity like a
bacillus
we see that this internal model there's
the exact same relationship the internal
model influences action
action influences the world plus it
influences itself
then the world influences observed
states sensory states
and those feed back into the internal
model of the bacterium
and so this is something that goes
beyond brains or neurons or neuroscience
this is something that has to do with
inference

and markov blankets any thoughts on that
one

uh i don't know maybe it will be late
important i think about

how markov blanket set up borders
with outside

of the system yeah we can

consider and work with it as a

real body and to define information

flows or something like this

yeah yeah so i think there we could go

to section 1.1

of the paper narratives as cognitive

schemas

um maybe i'm sort of

jumping ahead by just already talking

about the active inference perspective

on narratives

but when i think about uh narrative and

markov blankets

one thought is like a joke uh you know

three people walk into a bar

it's like it's already told you the who

and the what

it's told you who is entering what the

markov blanket

you know we're not talking about the

street outside the bar we're not talking

about

the uh office building down the road

we're talking about

the physical border of the bar

and we're talking about three people

entering that

so we still there's a lot of uncertainty

in that narrative but just by saying

three people walk into a bar um if the

next line of the story

is and then the chicken says it's like
it's a non-sequitur because
that is something happening uh in the
story
but it's not part of the actual uh
narrative it's not part of what was set
up
and so short stories or jokes or other
narrative retellings are like little
model systems
where we can explore how exactly are
narratives set up what are the inputs
what are the transformations that happen
during the narrative and then what are
the outputs
what do you think about that
yeah i think it's possible to consider
this perspective and um
maybe one thing i want to add
on this slide about
sensations and sensations and action
uh so if we can uh speak
about mental action and
a sensation like
mental representation of person for
himself for example
yep yeah so in this way
uh mental action
is it the same action or uh you need to
do a section with
here you need to do a section with uh
out of what
yeah for sure great great
great question and uh this is kind of
for example when people talk about
speech as a
motor behavior then it's like it all
works for when people are speaking but

then what about
the mental action of preparing speech or
of hearing voices but not vocalizing
them
so there's this question about
representation which is basically like
you said
is mental action action and then the
flip side of that is is mental
representation
sensation is that fair
yeah correct all right so here's what i
would say to that
um what we're experiencing as
conscious entities i mean not speak for
anyone else but what we're experiencing
is always our internal generative model
so that's why when we look out at the
world we don't see a blind spot
it's why the world appears to have
similar clarity
across the whole visual field have color
across the visual field etc
because we're experiencing our
generative model
and so when there's a uh mental
uh arising when there's awareness of
some
phenomena that isn't necessarily driven
by a change in the external
conditions so you're in the room it's a
dark room or it's a you know it's an
unchanging light and all of a sudden you
think about
being at the lake so that's an
experience
that you're having um it could be a
memory it could be a fantasy you could

put different words on it
but no sensory data entered you
to cause that thought to occur and so in
this
model of the markov blanket the
sensations are actually very very
specifically
the sensory information that's being
transmitted
through the primary senses so the
internal states
the dynamics of the internal states it
can be like it's like wow i'm bored in
this room
i wish i were at the lake now i'm
experiencing the lake
and so that's you can still have that
awareness and the representation of the
lake but that's actually not
happening through the sense uh thing so
again just there we're always
experiencing the internal
generative model uh so experiences are
not here
experiences are out here this is purely
about actual sense data
and then in the exact same way action is
actually referring
to um what is physically
embodied by the organism and so if the
organism is planning which strategy it's
going to use
policy planning and it's like okay i'm
going to wait 10 seconds and then as
soon as he goes left
i'm going to go right i'm going to run
past him that's
all part of the general model of the

world

it's part of the internal state and not

until the organism

actually acts is the action real

so mental experiences and mental

processes are all happening on the green

and the red is literally smell

taste sound vision and the blue is quite

literally motor behavior

does that categorize it or what do you

think about that

yeah cool inputs for me right now it's

more clear

good so um a few

let's go to section three narratives as

active inference

since um let's we can kind of get to the

real crux which is what does it mean

to take a active inference perspective

on narratives

and so one part that i thought that was

pretty

nice point was that this the second

sentence of

section three they say we suggest that

the capacity to build and share

narratives evolved so key it evolved

largely because it provides group

members with the ability to engage in

cooperative action by avoiding

unexpected states of

uncoordinated or contextually

inappropriate action

eg this is how we do things here not in

other ways

and so you reduce uncertainty for other

people

in a cultural context when you drive on

the correct side of the road
it would lead to a lot of uncertainty if
every single time
people had to come to a sort of
just-in-time decision about which side
of the road to drive on
and so norms whether they're implicit or
explicit
norms are really interesting to think
about
in the sense of reducing uncertainty
and so you can reduce uncertainty about
somebody's um
behavior in various ways and they
suggest that this leads to an account
of the conservative aspects of cultural
practice
or the traditional components really
just the idea that anything is
preserved is conserved or held as
traditional from the past
and that ensures that human agents that
are in culture
that way are able to reduce their
uncertainty when working with each other
what do you think about that or how does
that
seem to you too
uh for me it's interesting to address
from ideas of niche construction
and really from functional side it's how
you describe it
it's reducing uncertainty
and provides ways to be more effective
in communication
and our kind of
stuff doing in collective way
yeah so that's that's one on page 19.

that's just
one nice uh idea about how this
reducing uncertainty perspective of free
energy principle
rather than maximize reward it
contextualizes a lot of traditions
so for example it's hard to understand
how some cultural traditions
or cultural norms are simply about
maximizing reward because it's like
reward for whom
or how is it calculated or how is it
implemented
but rather when we switch out the thing
that has to be
done by culture as not just maximizing
success or reward
but actually just maximizing um
the precision of the action it makes a
lot more sense
so then at the end of page 19 this is a
second uh
area where narratives and active
inference are gonna
intersect they write coherent self
narratives
provide individuals with a mean to
maintain a stable course of action
pertaining to own particular goals and
experiences
which limits the occurrence of
unexpected states and so that first
account of uh cultural uncertainty
reduction
is something that's happening at the
level of like a population or a group
and this second uh sense of narrative is
more like something that somebody tells

themselves

as a means to maintain a stable course
of action so for example

i'm gonna go ride my bike to get some
batteries because my flashlight doesn't
have any batteries

and i'm riding to the store and then i
get a flat tire so now i need to walk my
bike to the bike store

but after i go to the bike store then

i'll be able to get those

batteries so i can go to the flashlight

it's like

it makes sense because that story
about i that story about what has to get
done

it's very flexible but it also prevents
the occurrence of unexpected states and
when unexpected states do occur

like getting a flat tire it can be
incorporated

into my narrative about my day instead
of

causing me to freeze or halt like a
computer if there's an unexpected
exception might just freeze or crash
but informational foragers real

evolutionary creatures can't do that

so that's one other very very personal
uh

way that narrative matters and then

just on page 20 furthermore narratives
are potential generators of innovation
and change and this is sort of combining
those previous two

the collective narrative precision uh
increasing components and then the
individual

precision increasing components of
narrative and then this is sort of
bringing them together
and saying that by doing active
inference
like search of imagining of exploring
imagining new possibilities in response
to or anticipation of adaptation
can reduce or uncertainty and so for
example if we say
we're going to reduce our uncertainty
about the future
of the climate by doing this and that
kind of modeling
and even though this or that weather
pattern hasn't
actually occurred we should build a new
wall here
and we should prepare this field for
growing this crop
so that's the way in which using
predictive action
um or acting in anticipation of
environmental changes can reduce
your uncertainty and that's a story that
you're telling about the future it's a
narrative
um any other
thoughts there um
i want to say is for self narratives
that
actually people spend most of their time
living in their self narratives and
as for really feeling
themselves the goods they always
creating some kind of
unreal stories and uh
really going sometimes going in the

wrong way mentally
yeah it's a one of the
the tenants about what leads to an
organism surviving
is the organism has to have a an
optimistic uh future model like it
the organism has to have a world view
where it sees itself existing in the
future
i mean why sleep if there's no tomorrow
or why eat if there's no tomorrow or
you know all these things and we see
that kind of survival reflex playing out
when
somebody's being held under water their
body will struggle because it's not
part of the expected sensory inputs to
be drowning
so on one hand we see this fundamental
struggle
for life which is deeply optimistic but
then at the same time
that goes awry when people sort of uh
end up licensing themselves to be the
judge jury and executioner so to speak
of the world and then they're the star
in their movie and they're walking
around the world and
everything is only mattering to the
extent it makes sense to them or it
benefits them
so having this uh personal level
narrative
is uh sometimes at odds with the
collective narrative
that's really a huge area of tension and
that
makes me think a lot about this

multi-level markov blanket

you know here here's us you know here's

one person let's just say on the left
side

here's stories coming in narratives

coming in memes coming out

so information in information out and

the hidden model is

someone's political beliefs or their

beliefs about lord of the rings

and then that person is actually taking

part in larger systems and so here's the

whole country or the whole

group chat or the whole team

and then that team also gets inputs and

gives outputs

and so that's coming back to your

original question

about what are markov blankets in the

context of narratives

and one element is markov blankets are

defined by what comes in and what comes

out

and that's informational in this case

and another answer or another aspect of

the answer

is because these markov blankets are

also defined by their

internal mechanics or at least their

internal dynamics

so dynamics would be referring to the

patterns that are just simply observable

and the mechanics would be the actual

rules that lead to those dynamics

arising

we can think what are those rules and

one example about where the rules

um could be a little different let's

just go back to that joke about three
people in the bar
we go three aliens walk into a bar on a
world
where gravity is reversed so it's like
i've put this extra rule
into the narrative and now other stuff
is going to happen in the narrative
that locally is going to make sense
because
i've also specified this other mechanism
that's going to be relevant
but without specifying that uh that
mechanism
then uh it's assumed
that the rules of gravity are regular
and actually
another funny thing and there's been
some work on this is with humor
and optimal surprise like simply to be
insulting is not funny as well as to say
what everyone else says is also usually
not funny but it's on that edge of chaos
where comedy and uncertainty and people
some are laughing some are crying some
people are trying to stop themselves
from laughing
all these things happen due to surprise
and so
to think about this joke and again not
to get too over
analyzing of jokes but when the joke
ends with a twist
and it was like you know uh something
that wasn't necessarily specified the
beginning but there's a
there's a double entendre there's a pun
these things are all

humorous and what's humorous about them
is they deliver that optimal level
of novelty or surprise
so let's go to
section 3.1 cognitive schema
and active inference on page 20. and so
this is going to be drilling down
a little bit into the cognitive schema
so this is internal
to someone's brain these are the uh the
cognitive
patterns that one person has
on board that are related to narratives
that
has to do with active inference and so
they basically write about how
having specific narratives of events
like
i was going to the store and then this
happened and then this happened or
i went to college and then this happened
those events
organize your life they org they give
meaning to what you have to do when you
have to wake up and not just like a
you know theological meaning like
literally it just gives
sense to your day it tells you which
direction to head out of the house
and then secondly there's this
meaningfulness that arises
from uh general events and collective
narratives
and they basically say that what we can
do and
also i'll just point out that those two
levels of analysis
the cognitive and the collective those

are referring back to when they said
that the issue with the field
is that the literature on narrative
focuses
on either one or the other so
psychologists will focus on
the internal cognitive schema of
narrative and the anthropologist well
focused on the cultural
researcher will focus on the collective
narratives
um and they're going to try to use
active inference to bridge that gap
and so basically they say on the quote
that goes from 20 to 21
um what explains these associations why
these individual collective narrative
roles
are adaptive in an evolutionary sense is
basically because they facilitate the
process of active inference in page 21.
that is the capacity of narratives to
represent the specificity of events and
to represent multiple events
in a meaningful and coherent way helps
individuals to more accurately
predict future events and therefore
respond to changing circumstances in
their lives
so that's pretty key what do you think
about that
um
as it's stated about active inference
that here in paper but it's first of all
about
adaptive behavior and
for sure this part is about how it's
work in real life

for example and people are rarely act
like this always living in some stores
and describe
for them for themselves future where
future possible events and
in terms for example engineering again
you first need to
create a description of future system
and how it will work
in future so it's way
yep yes and um and just at the end of
that

section on the bottom of 21
where they say in short having a
narrative or a repertoire of narratives
furnishes inference with a simple and
efficient way to account for sensory
evidence at hand
and what that allows uh organisms to do
is to then provide hypotheses that
enable inference to the best prediction
so you're telling the story and it's
like okay so there i was i had the
flashlight
and the batteries and my bike was
working and then
you can actually make hypotheses even if
you knew very little about the person
you could then say well what do you
think would happen if they found out
that the batteries were dead
oh they might go back to the store
because remember they needed the
batteries for the flashlight
or what would happen if this other thing
happened so
the narrative allows temporally related
events

to um become meaningful
and in the context of artificial
intelligence and in big data
this is so important because the
data overload means that let's just say
that you had somebody who
was they had agreed uh consented to
provide you all their data
so in the whole day you have their full
movement patterns
all their their smartphone and every web
search and all their transactions all
this stuff
you still might never be able to reduce
it to so
simple a story in just a few sentences
as i had to get batteries for my
flashlight but then my bike tire got a
flat
that is really really hard to extract
from just the gps data and the
transaction data
um because it's intermingled with so
many other
different components happening and so
narratives are
how we relate that sequence of events to
ourself
it's how we relate it to other people
and then the challenge is to generate
computational systems that can interact
at that level of analysis
um like you know okay google i want to
you know
get a haircut within three miles of
where i am
or something like that that's what we
don't have and that's why

the artificial intelligence community is
so interested in these topics
because potentially an act of inference
based artificial intelligence
uh both confusingly ai could
have a more uh effective role
in some of these questions so let's go
to
three one one narrative identity and
active inference
in this section they're gonna go a
little bit more into that first person
narration component and
on 22 they write coherent narratives
link together
aspects of our experience in ways that
decrease contradictions
amongst choices values and goals and so
the contextualization of narrative is
like
well it was really important to me to um
spend time with my family
but then for work i had to go away for
two days
and so that narrative is what allows
that person
to have two things uh that are otherwise
contradictory which is
well if families are important and work
is important then
what are you gonna do but the way that
that's narrated
it ends up not being that big of a deal
and so
it isn't that big of a deal because as
they point out
coherent narratives can reduce the
dissonance amongst the complex and

overarching expectations we have about
our lives
especially with relationship to future
states
so that's kind of i think one very
interesting
uh topic
anything there or yvonne any questions
or thoughts
uh and again again i want to put
about personal and uh
fun functionally for coherent narratives
that people in their in their
lives uh usually not action
we're trying to create a new narratives
way
why we don't uh do any action
and uh when we speak about
action it's about learning and
learning it's about updating your model
and it's
looks like it's quite difficult for
people stop to create
new narratives just not going to
do some actions yep
very very true um two quotes that that
really makes me think about on 22
about two-thirds the way down says self
and identity narratives are framed
against a background of cultural norms
and values
so they also can reference alternative
narratives that provide explanations for
non-normative
unusual inconsistent or untoward events
in the individual's life trajectory
so that'd be something like you know
when i was growing up

everybody uh was drafted into the army
but then because of this situation i
didn't have to
go so that explains why the individual
was doing something
explains why they had the sensory
experience of being at home for a
specific interval of time
but it also juxtaposes it against a
collective
potentially normative idea
and so this has a lot to do with humans
seeking to differentiate themselves
from each other is one tendency we want
to be unique in some ways but also
there's other areas where
uh we want to fit in and so there might
be cultures where it's
valued more to be fitting in or not
there might be
different dimensions where we fit in
different dish dimensions where we don't
that's kind of like
the idea of intersectionality and then
on the page 23 where they're talking
about
meaning making and sense making and i
i totally agree with that what you're
talking about
with most people being in a non-active
state the sensations coming in the you
know
media coming in or the internet coming
in
it's just updating their internal states
updating updating updating
but it doesn't seem to be reflected that
heavily in action

sometimes and then they write on the
middle 23

the meaning making process is an
adaptive response
to the occurrence of unexpected
circumstances when expectations are
broken

an imaginative reorganization of our
representations of ourselves and our
environment

is needed um but unfortunately if
somebody

sometimes gets a statistic or a fact
that is not consistent with their
worldview

then that's where we see logical
fallacies like oh ad hominem or that
that person's just

trolling or they're just uh you know
trying to frustrate me or something
because it's uh it's always possible to
include into your view just the idea
that that one fact is wrong
or irrelevant rather than actually move
into this totally new worldview because
shifting the deeper attendance of your
worldview

is uh anxiety producing and so then
also just to what you said about niche
they talk about how
uh before a new sense of familiarity and
coherence

i.e a new niche that's like a narrative
niche can be reinstated
exploration of new meaning epistemic
foraging

must occur in order to generate tests
and discover new adaptive inferences

so that's kind of a cool idea
let's go to uh 3.12
event narratives so this uh section 3.12
which is
a good length uh we can kind of just
skip to the summary
on 27 of this section where it says
event narratives are ideally suited for
active inference because of their
ability to represent past and future
events respectively in segmented event
sequence
and that provides a pool of uniquely
structured information that helps
predict future events
and structures these inferred in the
form of cohesive
future events so um
it it it seems like a very basic
point to build the paper towards which
is that
narratives help us recognize where we
are
recognize who we are personally and
culturally
recognize where we've been and then make
predictions about the future that
hopefully will help us reduce
uncertainty
these points are not major but i would
argue that it's actually quite
um encouraging and powerful that
these sorts of claims which usually
would be the qualitative
uh end point of some sort of
psychological or cultural analysis of
narrative
rather those are the starting points for

us having a
quantitative analysis potentially to
analyze narratives
um but that this section we've been kind
of discussing
uh throughout so section 3.13
integrating the functions of cognitive
schema narratives
as in as active inference so that's one
of the
final sections before they turned to
storytelling
um and uh
that just is about what it says it's
gonna be
about any thoughts on that one like
that's kind of related to what we've
been discussing as well
where they say on uh 27
on the bottom future projections and
episodic memory
can be understood as mental action or
policy selection and perception
so that kind of relates to the idea of
mental action
yeah yep
and then just to kind of finish out the
uh body of the paper before we return to
a few more images and close out
the section three two storytelling
regimes of attention and deontic
cues so uh regimes
of attention are this idea that um
you're paying attention to something you
know when
the child is in school and they're being
trained to pay attention to different
sorts of cues and

what symbols mean this idea that
attention
and that attention as shaped by active
inference
uh as on 29 where there's a few
citations to
constant at all 2019 and constant at all
2018
that the narratives direct what is
salient to human agents
um so it'd be like you know three people
walk into a bar
and one of them has three legs it's like
one of those statements
is surprising the one about three people
walking into a bar is not
but the one about somebody having a
different number of body parts is
and so narratives allow us to very very
easily
draw on the tremendous amount of
information that we have in our
generative model of the world
and then use that to discover anomalies
in our observations
and deontic is a word that
means duty so related to
what somebody uh must or should
do and theref and so
we can think about narratives for
ourself and collective narratives as
being
deontic so when jfk
you know said ask not what your country
can do for you but what you can do for
your country
it's almost like instead of thinking
about the

united states of markham blanket and how
that is going to provide something for
you
as well you might want to think about
you as a markov blanket
and how you as a personal narrative
active inference machine
might also be playing a role in this
larger system
uh maybe he was thinking about free
energy maybe he wasn't hard to say
and then they kind of closed the loop in
um
that paragraph in the middle of 30 where
they write
um deontic cues guide the sensory motor
patterns through which agents engage
adaptively their niches
action thereby guiding the ways in which
agents
update their recognition model so for
example when my alarm clock goes off
and i go okay i have 30 minutes to take
a shower
get dressed bike to school because i
have to be sitting in this lecture hall
so that i can receive the sensory input
of the lecturer
so that i'll be able to update my model
about organic chemistry
so that when there's a final i'm able to
reduce my uncertainty about what the
answer is
so that i'm able to get this grade that
i want and so
the deontic cue is like the post note
that might be on my
alarm clock that says get up you have to

go to organic chemistry
but that ends up guiding the sensory
motor patterns
and ideally that is a adaptive
engagement
through action and then that leads to
differences in perception and learning
and then this relates to a lot of other
work um i think it might be the vessier
uh at all 2020 citation on page 31
which is the idea of thinking through
other minds which
is a more recent development in the free
energy
slash active inference cultural
perspective
so let's go just to their conclusion the
authors are going to summarize
what they've been discussing they write
we have proposed an integrative model of
the diverse functions of narrative as
tools for active inference
in ways that clarify their cognitive and
social uses so again the big point of
the paper
was that there's this cognitive internal
level and this social external level and
the literature
tends to focus on one or the other they
brought the big guns out active
inference and so they're going to
integrate
cognitive and social under the umbrella
of active inference so that's going to
give
internal coherence to the study of
narrative and
it's also going to give external

coherence because now
we can study narrative alongside other
informational and thermodynamic
processes
then they in the following sentences
recall a bunch of the specifics
they talk about how narratives are
useful for segmenting events
for uh building an information pool for
making future projections
for meaning making and sense making for
belief
updating for assimilating unexpected
information
for maintaining cohesive integrity of
our complex network of expectations
so as to minimize internal
contradictions as well as providing
expectations
to culturally common situations and
typical life trajectories
um and then they close by discussing
that idea of the
deontic narrative and the narratives
that are about how the world could or
should
be and then that situates our role
in the world and that actually shapes
our sensory motor action patterns
and that their last sentence is just
that the
importance and ubiquity of narratives uh
in
the various domains of adaptation and
sociality it should be
explained not by appeal to entertainment
or
simply information exchange or

communication only
but rather by phrasing this all within
the context of active inference
we get a very very powerful framework to
get internal
and external coherence for narratives
um then just sort of in closing i
thought i would
go to a paper from
march 31st to 2020 uh active inference
on discrete state spaces a synthesis
and just a little bit looking at how
uh active inference what what could be
the next steps and this is something
that
uh you know the three of us have talked
about quite a lot which is
this paper is awesome it's also a
qualitative
perspective that doesn't dig into
all the details that we wanted to see
resolved that's why
research is on the edge and so these are
just a few of the ways
in which active inference has been
specified and so we might begin to think
about
how narrative could play into some of
this stuff
so here we see that familiar diagram
with the brain
internal states action states external
sensory states
and then there's basically two things
that are happening
two optimization processes that are
happening at the same time
perception is the brain making sense of

sensory data

and action is the brain's selecting

policy

and what happens is the way that free

energy is minimized

given the sensory data two things can

happen you can either

make sense of the sensory data by

changing your worldview

or you can change your sensory data

by engaging in action and so it turns

out that by different

transformations of these mathematical

relationships

that are specified exhaustively in this

paper

we pull out a few key terms so here's

value

this is the discounted or the expected

utility

it turns out that the value u_m expected

value of an action

or expected value of a stimulus relating

to questions about optimal control and

cybernetics as well these questions

about value

uh actually arise uh

purely from thinking about uncertainty

through time

uncertainty through time is also related

to uh

surprise this is why on both of these we

see this log

p be a given m u_m and this negative f

and f and then u_m also

we're able to introduce the topics of

entropy self-organization

synergetics homeostasis and of course

model evidence this is the key um
uh term pebm that's represented in all
of these
so those are a few interesting topics uh
this is another visualization
that talks about how active inference
and markov blankets
have very specific other downstream
mathematical relationships again
not too much to discuss at this point
but these are just to put up
and then this is the brain and here
are the different variables in their
brain
represented visually so here are
for example the generative model and how
things are being influenced and how
voter
cortex is ending up select policy moving
to the motor cortex
moving out and then maybe we can just
close
on this slide which alex i think you
put in where we have two agents
or more and here for agent one
here's their regime of attention this is
their internal model what they're paying
attention to what is salient for them
and this is related to who they are if
they don't speak the language they can
pay attention to the sounds but they
can't pay attention to the meaning
if they don't know the area they can pay
attention to the words but
not really the connections agent one
engages in action
and that kind of dumps into the world
and the world

is full of epistemic resources cues this
is just
what is and then the world
uh ends up dealing out sensations to
both of these agents and then it feeds
back and so it's kind of like the
conversation
would be okay this person is listening
they speak it dumps it into the text
room or the
the jitsi video chat that's sensed by
agent n
agent n pays attention acts that goes
back into the chat room
and so there's like this infinity here
uh and that's
thinking through other minds that's how
narratives are shared this is really
where we want to go
is thinking at a really formal level
as well as an applied level how can we
bring this kind of clarity to action
sensation and collaboration how can we
bring that
to the system's perspective the
complexity
yep perspective and um i want to
add here we see a green zone about
epistemic resources and silence and
as we see from this paper narratives are
really
one of fundamental part of
human well-being and developments
and culture social and so on
so and if so much functions
in it we need to uh
see closer to things like about
epistemic

side of his narrative and silent silence
part of his narratives and trying to
find
ways to have some kind of
evaluation and qualification for such
sources
or narratives as well
agreed agreed and again with this idea
of like there's the internal
coherence and then the external
coherence we can return to that joke
it's like in the context of the joke you
can make a mark of blanket for the joke
where three people walk into a bar
that's literally the three people
walking into a barkov blanket
um probably there's three people walking
to a mark called blanket
they get lost um but then zooming out
there's also the cultural context that
the joke occurs within
uh the kind of context that makes
uh three people walk into a bar now
that's common
it's an american or it's an english joke
format you know this many people walk
into a bar
or knock knock who's there now
that might be a culturally specific
thing and so that joke is going to be
more
or less expected more or less funny to
somebody who's in that cultural context
and so the epistemic
resources and the salience have to deal
with
the stimulus in question like the joke
itself

but also it has to be taken into account
that because we're talking about agents
that are paying attention
the culture of the agent is going to
dictate what they find salient
there might be things that are in the
world that are
not salient because the uh individual
doesn't have the narrative that allows
them to
retain that salience um
any closing thoughts especially yvonne
i just i just uh have
remembered one russian joke
that's congruent with the english
three people walking the barriers the
bill goes
in the forest it's uh it's
[Music]
start with the beer goes in the forest
and
it has a lot of different ants
in this story exactly
and some of these cultural stubs like
these starters
for jokes it allows uh the
format for anything it's kind of like a
http
packet you know you have the header that
just says i'm an internet packet
that's like saying you know when you say
a bear walks into a forest
it's like i'm telling you a joke but you
can't just say
i'm going to tell you a joke right now
and so these are
signals that are extremely rich with
information

and just having a text-based natural
language processing
ai will totally miss out on that
cultural
context and then let's say you told like
a variant of the joke you know
um uh an ant walks into the forest
it's not a non-sequitur but it's not
really referencing a real ant walking
into a forest
and these are these overlapping levels
of cultural
jokes and meaning that are
really pointing the way to how rich the
study could be with active inference
so how
how do you think
what the first thing that
is uh people don't have
people of different culture don't have
to
to do not missing
the point of communication how can
what do they can what can they
[Music]
do first to
reduce miscommunicating
yeah good question and especially in a
digital
and international context it's so
important
that there's reduction of uncertainty so
i mean just really quickly a few things
would be
just at every possible opportunity where
there's any question about uncertainty
i feel like always getting the uh
consent from both okay we were just

talking about this topic
right and so just getting used to
agreeing
with each other and getting used to
uh altruistically or just politely
exchanging information starts to train
us
to work well together even when the
situation
is very tense so that's one part is
where we can be extremely accurate in
communication
we should be extremely accurate and then
on the technological side
i think it's important just to
communicate simply instead of using
think oh we'll just use video and use
body language and make sure that you're
really expressive with your hands
uh yeah the tone of voice and the hand
motions
even if they come across with no lag and
the person is watching your screen
different cultures might unpack
different hand motions or different
facial expressions
very differently and so i think that
when there's
differences in culture that lead to
challenges in communication
it's critical to really just focus in on
what you're actually trying to
communicate
and make sure that there's as little
uncertainty as possible about
that and then the cultural stuff
deal with it as it's needed and whenever
it's needed to deal with it

um absolutely it's important to deal
with it because nothing can be more
important
than helping the team become more
effective more precise
i think in digital it's more
simple to um
to make clear enough communication
between
email between two people because in
when we when we are talking in real life
we can't to add some
hashtag to uh
describe what what do we uh
[Music]
what do we mean
yep we can use a rich multimodal
features so we could be having this chat
and someone could say
i'm sorry i just could you spell out the
city that you mentioned
and then they can type it in the text
box or
somebody can share their screen and draw
something out things that couldn't
happen in person
i i agree these are all really important
and that relates to epistemic resources
so when the people who you're dealing
with are all in the same discipline and
all the same culture
maybe the epistemic resources are very
professional
but when we're talking about broad
spectrum
communication potentially the epistemic
resources and the salience cues
would be like using a color scheme

that's accessible for everybody
and using icons to demonstrate
where the settings are like having a
gear in the bottom right of jitsi the
three dots
having a gear that says setting that's
really helpful because it's kind of like
a universal icon for the settings
and then otherwise it could be a
situation where someone doesn't know
english
and they can't find the settings so then
how are they going to be able to find
the language uh option because they
can't even find the settings
section but with the gear it's um
it's a salience cue that goes beyond
culture
uh and so that is what helps people get
along
together and again going back to like
what are the actual
applications of this if we could
understand
how individuals in different cultures
are going to perceive different
narratives and what role they see
themselves playing in the narratives
it could really open the door to
extremely
interactive and authentic narratives by
people
that don't have to be situated as being
at odds
with some sort of evil system and it's
like
they're they're the righteous crusader
fighting against everyone else who's

dumb

or the system's corrupt and they're just

you know doing what they can

all these narratives that that uh hit

the individual versus the group

or versus specific groups lead to

adversarial game theory and so

not not that there's never going to be

competition in the world but potentially

by having these protocols for

interaction and for

communication just like we have

protocols for

trade we could have protocols for

information that are also healthy

yeah cool and as for protocols uh

and you said before about signals and i

think

if you speak about culture it's really

mostly

uh shared by narratives and uh if he's

considered culture not like a global

uh gentleman but a local so

each team could have

own culture and they define it

by narratives and it creates a niche

by that supporting by tools and the

tools

really could support such protocols

where some meta information

added to communication and communication

process

is also structured according to

some models from active inferences we

see

totally agreed and i think on a closing

note

this has been our first podcast live

stream

discussion for team kong so

it's a a landmark event for us

we're using jitsi for

live streaming to youtube we used google

slides

for the presentation the pre-print

servers psi archive

and the internet in general are what

allowed us to make these presentations

and um we can really just

see this as happening as just another

event we're observing

this live stream come to an end it's

part of our narrative of

having met in uh you know in service of

these ideas and with a lot of the same

interests and so yeah this was really a

great discussion and i'm looking forward

to

continuing these discussions yeah thank

you very much for your leading

discussion i think results are

interesting and

we will use it in future for sure

all right i'm going to terminate the

live stream now